

Deepfake Forensic Classification Report

Packet ID: fb417038-59f2-42ec-90bd-da610ff757ca **Date (IST):** 04-06-2026 20:11:03 IST

Subject: Shri Demo Politician

Media SHA-256:
b94d27b9934d3e08a52e52d7da7dae55cfe5f806fb7c93f00039b7c69b6c4220

Media classified as likely synthetically manipulated based on 11 forensic indicator(s), including 4 high-severity finding(s).

Category	Severity	Finding	Metric
Audio-Video Sync	critical	Audio–video lip movement is misaligned, consistent with synthetic dubbing or post-hoc voice replacement rather than naturally captured speech.	lip_sync_alignment_c 135.0 ms
Facial Manipulation	high	Spatial facial inconsistencies detected across frames suggest GAN or deepfake-based facial manipulation.	spatial_cnn_manipula 92.00%

3D Face Geometry	medium	3D face mesh landmarks are unstable across the temporal sequence, indicating inconsistent facial geometry typical of synthetic faces.	face_mesh_landmark 0.0750
Synthetic Voice	high	The voice track exhibits characteristics consistent with text-to-speech or neural voice cloning rather than a live recording.	tts_synthetic_probabi 88.00%

Voice Authenticity	medium	Pitch contour and spectrogram features do not match natural speech patterns for the apparent speaker, suggesting audio synthesis or splicing.	spectrogram_pitch_m 39.00%
Anti-Spoofing	high	Anti-spoofing neural network assigns low confidence to bona fide speech, indicating the audio is likely non-genuine or heavily processed.	anti_spoofing_nn_co 11.00%

Lighting Consistency	medium	Lighting on the face and background shifts unnaturally between frames, suggesting compositing or generative synthesis rather than a single capture.	illumination_inconsis71.00%
Shadow Geometry	medium	Shadow direction and softness are inconsistent with the scene light source, a common artifact in deepfake and face-swap videos.	shadow_geometry_in66.00%

Color Grading	medium	Skin tone and color grading do not remain consistent across frames; color boundaries and shadows appear imperfect relative to natural video.	color_grading_anoma59.00%
Temporal Stability	medium	High-frequency flicker and temporal instability detected at facial boundaries, consistent with frame-wise generative manipulation.	temporal_flicker_sco55.00%

Facial Boundaries	medium	Visible artifacts at the jawline, hairline, or face perimeter suggest blending of a synthetic face onto an original body.	facial_boundary_artif 62.00%
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This report is generated for court and authority review under the Bharatiya Sakshya Adhiniyam, 2023 evidentiary framework.